

Addition and multiplication of probabilities

Two rules, one tree, and one question to ask first: does the second event depend on the first?

LESSON 83–85

3 × 40 MIN

ALGEBRA 11, §31

IGX 24.1, 24.4

LESSON CLOCK

15'

30'

35'

35'

25'

10

One question first	15 min
The multiplication rule	30 min
Trees	35 min
Total probability	35 min
Reversing the condition	25 min
Homework	10 min

LEARNING OBJECTIVES

- Use the multiplication rule for independent and for dependent events.
- Compute a conditional probability from data, a table or a tree.
- Build and use a tree diagram, including without replacement.
- Use the total probability formula, and reverse a conditional probability.

TEXTBOOK

National	pp. 365–382
Cambridge	Core & Extended, pp. 620–628, 639–642
Workbook	Exercise 24.4

01

Explanation

One question first

BEFORE ANY FORMULA

Does the second event depend on the first? If not, multiply the plain probabilities. If so, the second factor is a conditional probability.

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

When A and B are independent, $P(B \mid A) = P(B)$ and the rule collapses to $P(A)P(B)$. So the general rule always works; the independent one is a shortcut.

SITUATION	DEPENDENT?	SECOND FACTOR
two throws of a die	no	$\frac{1}{6}$
two cards, with replacement	no	$\frac{13}{52}$
two cards, without replacement	yes	$\frac{12}{51}$
two components from one batch	yes	from the reduced batch

The multiplication rule in use

$$P(A \cap B) = P(A) P(B \mid A) = P(B) P(A \mid B)$$

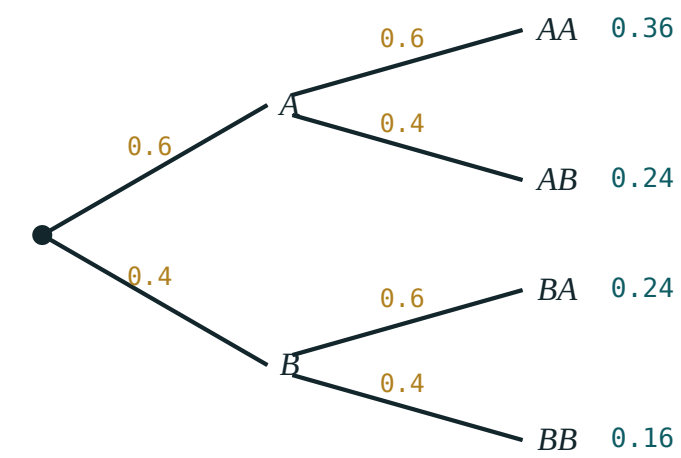
The rule can be entered from either end — a fact that will matter in the last section.

QUESTION	WORKING	ANSWER
two sixes in two throws	$\frac{1}{6} \times \frac{1}{6}$	$\frac{1}{36}$
two hearts, with replacement	$\frac{1}{4} \times \frac{1}{4}$	$\frac{1}{16}$
two hearts, without replacement	$\frac{13}{52} \times \frac{12}{51}$	$\frac{1}{17}$
two aces from 52	$\frac{4}{52} \times \frac{3}{51}$	$\frac{1}{221}$

THE DENOMINATOR DROPS AS WELL AS THE NUMERATOR

Without replacement, $\frac{13}{52}$ is followed by $\frac{12}{51}$ — **both** numbers change. Changing only the top is the standard error, and gives an answer about 2\% too large.

Trees



multiply along a branch · add down the ends

Multiply along a branch, add down the ends — the two rules in one picture.

THE TWO TREE RULES

- **Along** a branch: multiply (that is the multiplication rule).
- **Down** the ends: add (that is the addition rule, on exclusive paths).

Every fork's probabilities total 1; every set of end-probabilities totals 1. Two free checks.

Worked case. A bag holds 5 red and 3 blue balls; two are drawn without replacement.

PATH	PRODUCT	VALUE
RR	$\frac{5}{8} \times \frac{4}{7}$	$\frac{20}{56}$
RB	$\frac{5}{8} \times \frac{3}{7}$	$\frac{15}{56}$
BR	$\frac{3}{8} \times \frac{5}{7}$	$\frac{15}{56}$
BB	$\frac{3}{8} \times \frac{2}{7}$	$\frac{6}{56}$

They total $\frac{56}{56} = 1$. So $P(\text{same}) = \frac{26}{56} = \frac{13}{28}$ and $P(\text{different}) = \frac{30}{56} = \frac{15}{28}$.

Total probability

When the first stage splits the sample space into exclusive, exhaustive cases $H_1, H_2, \dots$, the probability of a later event is the sum over the branches:

$$P(A) = P(H_1)P(A | H_1) + P(H_2)P(A | H_2) + \dots$$

This is exactly “add down the ends”, written as a formula.

Example. Two factories supply lamps: 60% from F_1 with 2% defective, 40% from F_2 with 5% defective. The probability that a random lamp is defective:

$$P(D) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.020 = 0.032$$

THE BRANCHES MUST PARTITION

The cases must be mutually exclusive and cover everything: $P(H_1) + P(H_2) + \dots = 1$. If they do not, the formula gives a number that is not a probability — often greater than 1, which is the check.

Reversing the condition

Total probability answers “how likely is the effect?”. The reverse question — “given the effect, how likely is each cause?” — uses the same numbers the other way round:

$$P(H_1 | A) = \frac{P(H_1) P(A | H_1)}{P(A)}$$

The lamp again. A lamp is found defective; the probability it came from F_1 :

$$\frac{0.012}{0.032} = 0.375$$

So although F_1 makes 60% of the lamps, it accounts for only 37.5% of the defective ones — because its defect rate is lower.

$P(A | B)$ AND $P(B | A)$ ARE DIFFERENT NUMBERS

Here $P(D | F_1) = 0.02$ but $P(F_1 | D) = 0.375$ — a factor of nearly 19. Swapping the two is the commonest and most consequential error in probability, in examinations and outside them.

02 Worked examples

EXAMPLE 1 Two cards are drawn from 52 without replacement. Find $P(\text{both hearts})$.

$$\textcircled{1} \quad P(\text{first}) = \frac{13}{52} = \frac{1}{4}$$

$$\textcircled{2} \quad P(\text{second} \mid \text{first}) = \frac{12}{51}$$

Both numbers change.

$$\textcircled{3} \quad P = \frac{1}{4} \times \frac{12}{51} = \frac{1}{17}$$

ANSWER

$$\frac{1}{17} \approx 0.059$$

EXAMPLE 2 A bag holds 5 red and 3 blue balls; two are drawn without replacement. Find $P(\text{same colour})$.

$$\textcircled{1} \quad P(RR) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$$

$$\textcircled{2} \quad P(BB) = \frac{3}{8} \times \frac{2}{7} = \frac{6}{56}$$

$$\textcircled{3} \quad P = \frac{26}{56} = \frac{13}{28}$$

Add down the ends.

ANSWER

$$\frac{13}{28} \approx 0.464$$

EXAMPLE 3 Factory F_1 makes 60\% of lamps with 2\% defective; F_2 makes 40\% with 5\%. Find the probability that a lamp is defective.

$$1 \quad P(D | F_1) = 0.02, P(D | F_2) = 0.05$$

$$2 \quad P(D) = 0.6(0.02) + 0.4(0.05)$$

$$3 \quad = 0.012 + 0.020 = 0.032$$

ANSWER

0.032

EXAMPLE 4 A lamp is defective. Find the probability that it came from F_1 .

$$1 \quad P(F_1 \cap D) = 0.012$$

$$2 \quad P(D) = 0.032$$

From the previous example.

$$3 \quad P(F_1 | D) = \frac{0.012}{0.032} = 0.375$$

ANSWER

0.375

03 Interactive model

Draw the lamp tree on the board and label all four end-probabilities; the reversal is then a single fraction the class can see.

Along the branch, down the ends

Along a branch you:

add

multiply

subtract

divide

Down the ends you:

add

multiply

subtract

divide

Without replacement, the second denominator:

stays

drops by one

doubles

is 1

$P(A \cap B)$ in general:

$$P(A)P(B)$$

$$P(A)P(B | A)$$

$$P(A) + P(B)$$

$$P(B | A)$$

The branches of total probability must:

be equal

partition the space

be independent

be few

All end-probabilities of a tree total:

0

1

2

anything

$P(D | F_1) = 0.02$ and $P(F_1 | D) = 0.375$ shows:

an error

the two are different

they are equal

nothing

For independent events, $P(B | A)$ equals:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Conditional probability	Shartli ehtimollik	Условная вероятность
Independent events	Bog'liqsiz hodisalar	Независимые события
Dependent events	Bog'liq hodisalar	Зависимые события
Tree diagram	Daraxt diagrammasi	Дерево исходов
With replacement	Qaytarib	С возвращением
Without replacement	Qaytarmasdan	Без возвращения
Total probability	To'la ehtimollik	Полная вероятность
Bayes' rule	Bayes formulasi	Формула Байеса
Partition	To'liq guruh	Полная группа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The general multiplication rule is:

$$P(A)P(B)$$

$$P(A)P(B | A)$$

$$P(A) + P(B)$$

$$P(A) - P(B)$$

Without replacement makes the events:

independent

dependent

exclusive

certain

Total probability is the formula for:

along a branch

down the ends

the complement

the union

Reversing a conditional probability uses:

the addition rule

the same joint probability, divided differently

independence

a new experiment

A tree fork's probabilities total:

0

1

$|U|$

anything

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. P two sixes in two throws

ANSWER $\frac{1}{36}$

2. P two hearts, with replacement

ANSWER $\frac{1}{16}$

3. P two hearts, without replacement

ANSWER $\frac{1}{17}$

4. P two aces from 52 without replacement

ANSWER $\frac{1}{221}$

5. Bag of 5 red, 3 blue: $P(RR)$ without replacement

ANSWER $\frac{5}{14}$

6. Same bag: $P(BB)$

ANSWER $\frac{3}{28}$

7. Second factor after drawing one heart from 52

ANSWER $\frac{12}{51}$

Medium · the standard question

7 PROBLEMS

1. Bag of 5 red, 3 blue: $P(\text{same colour})$

ANSWER $\frac{13}{28}$

2. Same bag: $P(\text{different colours})$

ANSWER $\frac{15}{28}$

3. Lamps: $P(D)$ with $0.6/0.02$ and $0.4/0.05$

ANSWER 0.032

4. Same: $P(F_1 | D)$

ANSWER 0.375

5. Same: $P(F_2 | D)$

ANSWER 0.625

6. Three cards from 52 without replacement: P all hearts

ANSWER $\frac{11}{850} \approx 0.0129$

7. $P(B | A)$ when $P(A) = 0.4$, $P(A \cap B) = 0.1$

ANSWER 0.25

Hard · stretch and reason

7 PROBLEMS

1. A bag of 4 white, 6 black: P the second drawn is white

ANSWER $\frac{2}{5}$

2. Two bags: I has 3R 2B, II has 1R 4B. A bag is chosen at random, then a ball: $P(R)$

ANSWER 0.4

3. Same: $P(\text{bag I} \mid \text{red drawn})$

ANSWER 0.75

4. A test is 95% accurate; 1% of people have the disease. $P(\text{disease} \mid \text{positive})$

ANSWER ≈ 0.161

5. A die is thrown until a six appears: P it takes exactly three throws

ANSWER $\frac{25}{216}$

6. Three components each fail with probability 0.1 , independently: P at least one fails

ANSWER 0.271

7. Two cards from 52 without replacement: P at least one is an ace

ANSWER $\frac{33}{221} \approx 0.149$

07 Homework

Homework — 5 tasks

Draw the tree for every multi-stage question; label every branch before multiplying.

1. A bag holds 6 green and 4 yellow counters; two are drawn without replacement. Find the probabilities of two greens, of two yellows and of one of each.
2. Two cards are drawn from 52 without replacement. Find the probability that at least one is a king.

3. Machine A makes 70\% of the output with 3\% faulty; machine B makes 30\% with 6\%. Find the probability that an item is faulty, and the probability it came from B given that it is.
4. A die is thrown three times. Find the probability of at least one six.
5. Explain in two sentences, with the lamp numbers, why $P(A \mid B)$ and $P(B \mid A)$ are different.